

Project Code

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#include <iostream>

#include <string>

using namespace std;

// ----- PET CLASS -----

class Pet {
private:
    string name;
    int hunger;
    int happiness;
    int energy;
    // Hidden behavior counters
    int feedCount;
    int playCount;
    int neglectCount;
    int trainCount;
    string personality;
public:
    // Constructor
    Pet(string n) {
        name = n;
        hunger = 50;
        happiness = 50;
        energy = 50;
        feedCount = playCount = neglectCount = trainCount = 0;
        personality = "Neutral";
    }
    // ----- ACTIONS -----
    void feed() {
        cout << name << " is eating...\n";
        hunger -= 15;
        happiness += 5;
        feedCount++;
    }
};
```

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        updatePersonality();
    }

    void play() {
        cout << name << " is playing...\n";

        happiness += 15;

        energy -= 10;

        playCount++;

        updatePersonality();
    }

    void sleep() {
        cout << name << " is sleeping...\n";

        energy += 20;

        hunger += 10;
    }

    void train() {
        cout << name << " is training...\n";

        happiness -= 5;

        energy -= 10;

        trainCount++;

        updatePersonality();
    }

    void neglect() {
        cout << name << " feels ignored...\n";

        hunger += 10;

        happiness -= 10;

        neglectCount++;

        updatePersonality();
    }

    // ----- PERSONALITY LOGIC -----

    void updatePersonality() {
        if (feedCount >= 3)
            personality = "Lazy";

        else if (playCount >= 3)
            personality = "Friendly";

        else if (trainCount >= 3)

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        personality = "Intelligent";
    else if (neglectCount >= 3)
        personality = "Aggressive";
    else
        personality = "Neutral";
}

// ----- DISPLAY -----

void showStatus() {
    cout << "\n----- PET STATUS ----- \n";
    cout << "Name: " << name << endl;
    cout << "Hunger: " << hunger << endl;
    cout << "Happiness: " << happiness << endl;
    cout << "Energy: " << energy << endl;
    cout << "Personality: " << personality << endl;
    // Personality-based dialogue
    if (personality == "Friendly")
        cout << name << ": I love spending time with you! \n";
    else if (personality == "Lazy")
        cout << name << ": Can I just sleep...? \n";
    else if (personality == "Aggressive")
        cout << name << ": Leave me alone! \n";
    else if (personality == "Intelligent")
        cout << name << ": Let's learn something new! \n";
    else
        cout << name << ": I'm feeling okay.\n";
}

};

// ----- MAIN PROGRAM -----

int main() {
    string petName;
    cout << "Enter your pet's name: ";
    cin >> petName;
    Pet myPet(petName);
    int choice;
    do {

```

```
    cout << "\n==== EvoPet Menu ==== \n";

    cout << "1. Feed\n";
    cout << "2. Play\n";
    cout << "3. Sleep\n";
    cout << "4. Train\n";
    cout << "5. Ignore (Neglect)\n";
    cout << "6. Show Status\n";
    cout << "0. Exit\n";
    cout << "Enter choice: ";
    cin >> choice;

    switch (choice) {
    case 1: myPet.feed(); break;
    case 2: myPet.play(); break;
    case 3: myPet.sleep(); break;
    case 4: myPet.train(); break;
    case 5: myPet.neglect(); break;
    case 6: myPet.showStatus(); break;
    case 0: cout << "Goodbye!\n"; break;
    default: cout << "Invalid choice!\n";
    }

    } while (choice != 0);
    return 0;
}
```